

A Comparative Study of YOLO-Based Models for Multi-Class Ovarian Disease Classification

M. Sai Meghana¹, K. L. Anvitha¹, P. Pavitra Kumari¹, and B. Saranya Devi^{1*}

Department of Computer Science & Engineering, Amrita School of Computing,
Bengaluru,

Amrita Vishwa Vidyapeetham, India

`bl.en.u4cse23132@bl.students.amrita.edu`,

`bl.en.u4cse23130@bl.students.amrita.edu`,

`bl.en.u4cse23141@bl.students.amrita.edu`, `b.saranya@blr.amrita.edu`

*Corresponding Author: `b.saranya@blr.amrita.edu`

Abstract. Early and accurate diagnosis of ovarian disorders is highly inaccurate. Important, to make clinical choices and interventions. This will also be useful in improving patient outcome, in a quick and effective way. and make diagnosis of those disorders. It has been found that one of the most frequently used methods screening ovarian abnormalities is through the use of ultrasound imaging, because it is non-invasive and inexpensive, and can display images in real time. The ultrasound images produced to be used in the diagnosis of the same are however, generated. Difficult processes are given by difficult to interpret speckle noise. low contrast with the surrounding tissue, operator dependency, and there are close parallels of various kinds of ovarian abnormalities. A comparative analysis of three You Only Look Once (YOLO)-based. identification of multi-class ovarian disease using deep learning models. Multi-Modality Ovarian Tumor Ultrasound Multi-Mode Ovarian Tumor ultrasound images. (MMOTU) dataset [26]. YOLOv8-cla, YOLOv11-cla and YOLOv12-cla. were experimented under the same conditions, which gave fair chances of experimentation. comparison of each model. YOLOv11-cla had highest top-1 test. The accuracy of 77.08%, as compared to both YOLOv8-cla and YOLOv12-cla.

Keywords: Ovarian Disease Classification, Ultrasound Imaging, Deep Learning, YOLO, Medical Image Analysis

1 Introduction

Ovarian disease is a heterogeneous group of ovarian disease (benign and malignant) known to have a significant effect on the health of women worldwide. Correct diagnosis of ovarian disease at an early age is significant to improve the survival rate and treatment outcome because the late diagnosis of the malignant disease in particular may lead to severe complications. Nevertheless, ovarian disease is hard to identify at an initial stage because clinical manifestations and

general appearance on the nature of imaging of the various ovarian pathologies are similar. Ultrasound imaging is therefore the main diagnostic method utilized as the first-line in women who are suspected of having ovarian disease due to its low costs, non-invasive and presents real-time images of the anatomy of the ovary.

Although ultrasound has a number of strengths on the diagnosis front, it has numerous challenges. These include problems in the image quality of the included images that include speckle noise, low contrast resolution and dependency on operators that is causing inconsistency in how images are classified by different radiologists; the similarity in the different categories of ovarian disease makes it hard to classify the various images correctly even in experienced clinicians; the above issues are indicative that automated and computer aided diagnostic devices are needed to assist clinicians.

Recently, a lot of advances have been made in applying deep learning methods to medical images, in particular, disease detection and classification. Convolutional Neural Networks (CNNs) were proven to be highly efficient tools; yet, the majority of the existing state-of-the-art models are highly computationally-intensive, thus, they are not applicable to real-time clinical use. Originally founded on real-time object detection, the YOLO (You Only Look Once) architecture has recently been applied to image classification problems; the architecture in particular is highly effective due to the efficiency with which the model is able to extract salient features of an image at significantly reduced computation costs as compared to CNN-based models. Thus, YOLO-based models would be the best applicants to be implemented in a clinical setting since both speed and accuracy are equally important.

This paper compares three out of these benefits encouraged by these benefits. Multi-class ovarian disease classification based on YOLO classification models. with MMOTU [26] of ultrasound images-YOLOv8-cl, YOLOv11- cl and YOLOv12-cl. To compare the accurateness of the models well, strength, and computational efficiency, all the models were trained and evaluated. at the same experiment conditions.

The key contributions of the paper are the following:

- Comparative study of various YOLO-based multi-class architecture classification of ovarian disease using ultrasound data (YOLOv8-cl, YOLOv11-cl, and YOLOv12-cl), and the performance of every architecture to categorise the ovarian disease is thoroughly investigated.
- Standardized techniques of assessing and training every architecture to en-off all architectures would be comparative with each other under standard. conditions.
- Evaluating the model using the MMOTU ovarian ultrasound data. [26], and addressing various data issues such as class imbalance and image similarities among classes of ovarian disease.

The remainder of the current paper will be structured in the following manner: Literature review on the topic. subject of automated classification of ovarian

by ultrasound imaging (Section 2); Characterization of the dataset and preprocessing methodology of this study. The experimental methodology and methodology validation (Section 3); Description of the experimental methodology and methodology validation (Section 4); Results of the models under assessment (Section 5); Visual validation of the execution of the suggested results of each of the assessed models (Section 6); Limitations and future recommendations (Section 7).

2 Literature Review

Ultrasound-based technologies of automated tumor diagnostic in the ovary have been researched widely in the last twenty years. Early automated methods primarily image processing and manual feature extraction that were manually processed and manually features and were mainly processed. incapable of representing the complicated shape, size, and texture patterns of ovarian tumors. Due to the development of deep learning, this has been significantly enhanced. ultrasound image interpretation in the performance and dependability of ultrasound images. ovarian tumors.

The proposed study by Dai et al. [1] created and tested a deep learning pipeline to diagnose ovary masses using ultrasound screening in a multicenter retrospective study, showing a good generalization across sites of clinical practice. Su et al. [2] proposed a multimodal deep learning model with clinical-ultrasound images. data which has better ovarian cancer diagnosis than image only. approaches. Du et al. [3] provided a deep learning radiomics based on ultrasound. model of distinguishing benign, borderline, and malignant ovarian tumors in a multi-class classification study, which demonstrates the importance of combined radiomics. and deep learning features. Li et al. [4] created a deep learning model system. to diagnose and treat adnexal masses, proving to be reliable. response in different tumor groups.

Transfer learning has extensively been embraced in studies of ovarian ultrasound. to solve the problem of the scarcity of annotated medical data. Wu et al. [5] applied the deep convolutional neural networks to classify various histopathological results in reporting logic types of ovarian tumors in ultrasound images. across classes. A systematic review of deep learning was carried out by Olosunde and Onuiri [6]. early-detection ovarian cancer learning models based on deep learning ultrasound. agnostic tools, overarching architectural trends and challenges. Tiwari et al. [7] suggested attention-based lightweight deep learning architecture to the. Ultrasound images Classification of polycystic ovary syndrome based on ultrasound images, focusing on. effectiveness and understandability. Patil et al. [8] came up with an intelligent computer-aided diagnosis system, in the classification of ovarian masses with the help of a machine. learning style, the hybrid of handcrafted and classification models.

Liu et al. [9] examined the combination of radiomics and deep learning features. Combining features in pattern classification of ovarian tumors, it is seen that combining features does not enhance the classification process. learned and handcrafted

enhances the accuracy of discrimination. Radhakrishnan et al. [10] used deep learning to further advance the diagnosis of ovarian cancer. explainable AI, which provides a multiclassification method with Grad-CAM visualizations to improve clinical transparency and trust.

Recent research has been switched towards multi-classification of ovarian disease deep learning architectures. Rani and Arrama [11] proposed a new deep learning system of multi-class ovarian tumor detecting. The accuracy of models with the use of ultrasound imaging is shown to be improved across benign, borderline and malign categories. The deep develops by Kolekar et al. [12]. EfficientNetB2 with an ovarian cancer classification model based on learning. attention mechanism, classification improved feature extractions and classification. performance. EfficientOvaNet was proposed by Parekh and Desai [13], and it is an efficient. Multiclass classification of benign ovarian cysts using deep learning model with ultrasound images, competitive accuracy, reduced computation. cost.

Garcia-Atutxa et al. [14] suggested the early-fusion hybrid CNN-transformer models to use multiclass ovarian tumor ultrasound classification, use complementary merits of convolutional and self-attention processes. Sundari et al. [15] used a CNN model with transfer learning to merge ultrasound of integrative. demonstrating sound and biomarker-based diagnosis of polycystic ovarian disease. indicating that it is the advantage of integrating imaging and clinical biomarkers. Zhou et al. [16] performed a comparative study involving a large set of samples across multiple centers and incorporating deep learning. Conventionally, ovarian tumor classification and diagnosis through radiomics analysis, strengthening the effectiveness of hybrid solutions with a variety of data sets. Bornhauser et al. [17] proposed a recently curated ultrasound dataset to diagnose using AI. of ovarian disorders, and offers indications of how a model performs on a new data. source.

Sajjan et al. [18] suggested an improved frame of ovarian cyst classification. train on a deep learning model, and report the improvement of the sensitivity and specificity. A deep learning method of classification was introduced by Giourga et al. [19]. Differentiating benign, malignant, and borderline tumors in ovaries with the help of convolutional. data augmentation neural networks with generative adversarial networks. mentation. Zhang et al. [20] used deep learning to differentiate between primary. histopathology images of primary and metastatic mucinous ovarian carcinoma, presenting high competence in discrimination. Gülmez [21] conducted a systematic literature review of deep learning methods of ovarian cancer detection, dealing with the issues of clinical translation and the future. Saha et al. [22] presented PCOS-proposed PCOS fresh, a hybrid deep feature fusion network. sification based on ultrasound images, which feature-level is effective. fusion strategies.

Most recently, deep learning architectures built on the YOLO have displayed high prognosis of real-time medical images analysis because of their computationally. efficiency and speed. The research by Devi et al. [23] proved that YOLO- is efficient. object detection in complex real-time scenarios using based deep learning techniques, determining the appropriateness of YOLO-based applications to time-sensitive visual analysis tasks. The traditional methods have also been

used in detecting ovarian cancer. machine learning methods. Inture et al. [24] exploited random forests of ovar. ian cancer detection and accuracy prediction, to a reasonable performance; but these are manual feature engineering methods and have lower gen- comparable to deep learning methods. The MMOTU dataset [25, 26], a multi-modality ovarian tumor ultrasound image dataset, has acted as a major one. reference point of measuring classification and segmentation models in this field.

Table 1: Summary of Existing Deep Learning Models for Ovarian Tumor Classification

No.	Authors	Models Used
1	Dai et al. [1]	CNN with transfer learning (ResNet-based)
2	Su et al. [2]	Multimodal DL (CNN + clinical data)
3	Du et al. [3]	Deep learning radiomics (CNN + radiomics)
4	Li et al. [4]	Deep CNN (ResNet, DenseNet)
5	Wu et al. [5]	Deep CNN (multi-histologic classification)
6	Olosunde and Onuiru [6]	Review: CNN, TL, attention-based DL
7	Tiwari et al. [7]	Attention-guided lightweight CNN
8	Patil et al. [8]	CNN + SVM, machine learning ensemble
9	Liu et al. [9]	Radiomics + deep learning feature fusion
10	Radhakrishnan et al. [10]	Explainable CNN + Grad-CAM
11	Rani and Arrama [11]	Novel deep CNN (multi-class)
12	Kolekar et al. [12]	EfficientNetB2 + attention mechanism
13	Parekh and Desai [13]	EfficientOvaNet (lightweight CNN)
14	Garcia-Atutxa et al. [14]	Hybrid CNN-Transformer (early fusion)
15	Sundari et al. [15]	Transfer learning CNN + biomarkers
16	Zhou et al. [16]	Deep learning + radiomics (multicenter)
17	Borna et al. [17]	CNN on novel ultrasound dataset
18	Sajjan et al. [18]	Enhanced deep learning classification
19	Giourga et al. [19]	CNN + GAN (data augmentation)
20	Zhang et al. [20]	Deep CNN (histopathology-based)
21	Gülmez [21]	Systematic review: DL for ovarian cancer
22	Saha et al. [22]	PCOSFusionNet (hybrid feature fusion)
23	Devi et al. [23]	YOLO-based real-time detection
24	Inture et al. [24]	Random Forest classification

Despite the major progress made in this field, the majority of the current work still utilizes the traditional CNN architectures or highly computational complex models. A systematic evaluation of recent YOLO-based classification models of multi-class ovarian. The ultrasound image-based disease detection is hardly studied. This gap is the major driving force of the current research that is intended to assess and evaluate. compares numerous models of YOLO-based classification with a common experimental design to determine the most appropriate architecture to use in order to achieve accurate and efficient architecture. classification of ovarian diseases.

3 Dataset Description

The MMOTU dataset[26] is a publicly available ovarian ultrasound dataset. used in this study. The data set includes ultrasound images that were taken during.

Table 2: Class-wise Distribution of the MMOTU Dataset

Class Name	Number of Images
Normal Ovary	336
Teratoma	336
Chocolate Cyst	267
Simple Cyst	219
Theca Cell Tumor	104
Serous Cystadenoma	88
Mucinous Cystadenoma	66
Total	1416

routine clinical and has also been extensively applied to the classification of ovarian tumors classification research.

Imperfect competition This dataset has been widely used in studies on the classification of impeccable competition. ovarian tumors and diseases. The data that was received at first consisted of a number of ultrasound images of 8 groups of ovarian tumors. Of these classes, the high-grade serous type was abolished because of the following reasons: of importance. unequilibrium of the samples in this category, very high volumes of disparities between doctors and interpreting the images, lack of sufficient credible samples to prove its features appropriately. Once this set has been eliminated, a set remains. of datasets have 7 classes (chocolate cyst, serous cystadenoma, teratoma, theca cell tumor, simple cyst, normal ovary, and mucinous cystadenoma).

The data set that results after filtering of classes and preprocessing includes a total. of 1416 ultrasound images. The the distribution of the class-wise is given in Table 2. MMOTU dataset, where the ratio between the classes is very high as there is a significant imbalance in favor of either Normal Ovary or. Teratoma the most represented classes (336 images each), and Mucinous Cystadenoma is the sample with the lowest number of samples (66 images). This was an imbalance that was corrected. with data augmentation schemes in training. The ultrasound images were modified to 224x224 pixels, and altered to. converting grayscale to RGB format, whereby they could be reformatted to be used in. Deep Learning Models. As a measure of assisting the model in learning more in training, data augmentation algorithms; such as, but not restricted to, image rotation. around their centre-point, turning them upside-down and downside-up, and altering the brightness/contrast of image, and introduction of speckle noise were done. To ensure that there are equal distributions of classes in each of the datasets. A stratified trainvalidation-test split is used as a trainvalidation-test split with the purpose of training, validation and testing. methodology was applied. This led to the split of the entire dataset into There are 988 training images, 284 validation images and 144 test images.

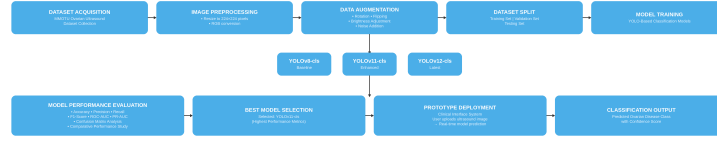


Fig. 1: Flowchart of the proposed YOLO-based ovarian disease classification framework.

4 Methodology

The usefulness of the YOLO-based deep learning models in multi-class ovarian. Evaluation of disease classification based on ultrasound images was assessed based on the controlled method comparative framework.

The ovarian ultrasound dataset used is MMOTU dataset [26] which is made publicly accessible. used in this study. The data set comprises of ultrasound images that were recorded. during.outline clinical, has also been widely applied to the classification. of tumours of the ovary. classification research.

Fig. 1 shows the general workflow of the suggested system, which involves There are five steps in total: data acquisition, preprocessing, data augmentation, model. performance evaluation, training and performance evaluation. The flowchart shows the process of raw ultrasound.graphs of the MMOTU database are analyzed with each step prior to conclusion.the models based on YOLO produce classification output.

Initially, the ovarian ultrasound data of MMOTU [26] was used. pruned down to allow it to be compatible with deep learning models. Input dimensions were scaled to bring all the ultrasound images to 224 RGB and 224x224. x3 pixels. The technique of data augmentation that was employed was rotation, horizontal and vertical flipping, brightness adjustment and addition of noises to. enhance the generalization of models and eradicate the issue of imbalance of classes.

This was followed by the stratification of the dataset into training, validation and testing. subsets to preserve the frequency of classes in all the subsets. This guaranteed the objective evaluation of model behavior. To compare, three YOLO-based classification architectures, which include YOLOv8-cls, YOLOv11-cls, and YOLOv12-classification architectures, YOLOv8-cls, YOLOv11-cls and YOLOv12- Experimental conditions under which were trained, like the choice, were the same with cls. of optimizer, the learning rate, and the batch size, the loss function, and the number of epochs.

YOLOv8-cls was taken as a baseline, which will be used with equal performance and. computing efficiency. Architecture additions were made to YOLOv11-cls in order to improve feature aggregation without sacrificing on the lightweightness of the archi- architecture, and that is applicable in real-time. YOLOv12-cls was a further in architecture which was meant to explore the possible of more dis- Medical imaging data learning criminative features with YOLO classifiers. Performance measures of the trained models were applied using multiple measures, and classification accuracy, validation loss, precision, recall, F1-score, Re-

Table 3: Comprehensive performance metrics and class-wise evaluation of YOLO-based models on the MMOTU test dataset

Model	Acc.	AUC	PR-AUC	Class	Prec.	Rec.	F1
YOLOv8	70.14	0.9242	0.6864	Chocolate Cyst	0.6389	0.8519	0.7302
				Mucinous Cystadenoma	0.5000	0.1429	0.2222
				Normal Ovary	0.8235	0.8235	0.8235
				Serous Cystadenoma	0.6250	0.5556	0.5882
				Simple Cyst	0.6667	0.4545	0.5405
				Teratoma	0.6905	0.8529	0.7632
				Theca Cell Tumor	0.7143	0.4545	0.5556
				Macro Avg	0.6655	0.5908	0.6033
YOLOv12	53.47	0.7651	0.4340	Chocolate Cyst	0.4130	0.7037	0.5205
				Mucinous Cystadenoma	0.0000	0.0000	0.0000
				Normal Ovary	0.6047	0.7647	0.6753
				Serous Cystadenoma	0.0000	0.0000	0.0000
				Simple Cyst	0.5517	0.7273	0.6275
				Teratoma	0.6400	0.4706	0.5424
				Theca Cell Tumor	0.0000	0.0000	0.0000
				Macro Avg	0.3156	0.3809	0.3380
YOLOv11	77.08	0.9351	0.7691	Chocolate Cyst	0.6857	0.8889	0.7742
				Mucinous Cystadenoma	0.6667	0.2857	0.4000
				Normal Ovary	0.8529	0.8529	0.8529
				Serous Cystadenoma	0.5833	0.7778	0.6667
				Simple Cyst	0.7059	0.5455	0.6154
				Teratoma	0.8750	0.8235	0.8485
				Theca Cell Tumor	0.8182	0.8182	0.8182
				Macro Avg	0.7411	0.7132	0.7108

Table 4: Overall Macro-Averaged Precision, Recall, and F1-score of YOLO-Based Models on the MMOTU Test Dataset

Model	Precision	Recall	F1-score
YOLOv8-cls	0.6655	0.5908	0.6033
YOLOv12-cls	0.3156	0.3809	0.3380
YOLOv11-cls	0.7411	0.7132	0.7108

ceiver operating Operating Characteristic Area Under the Curve (ROC-AUC), Precision-Recall Area Under the Curve (PR-AUC). These steps provided elaborate intelligence, classification on the stability of the model, the performance of the models per class and performance.

Table 3 shows the overall class-wise performance measures of all in case of data set imbalance, three YOLO models on MMOTU test dataset and per-class precision, recall and F1-score, and the values of overall accuracy and AUC.

5 Results and Discussion

We have compared the performance of YOLO based classification model with MMOTU. A fixed stratification train-validation-test ovarian ultrasound dataset [26] . split. The results in Table 4 show the macro-averaged precision, recall and F1-score of all three models, which proves that YOLOv11-cls has the highest overall performance in all three metrics that the macro-averaged F1-score is 0.7108, compared to YOLOv8-cls (0.6033) and YOLOv12-cls (0.3380). Each of the models was subjected in the laboratory of the 50 epochs to train in the same manner. make comparative objective analysis on models. Top-1 was among the

evaluation measures. Accuracy of Classification, Macro-Averaged ROC-AUC, Macro-Averaged PR- AUC, and class-wise metrics containing Precision, Recall and F1-score, all of which are less than one. as are summarised in Table 3.

5.1 Overall Performance Comparison

YOLOv11 has the best overall test accuracy of as indicated in Table 3 . A higher percentage 77.08% than YOLOv8 (70.14%) and YOLOv12 (53.47%). The per- The class separability of YOLOv11 is better illustrated by its performance, and the macro-separability is higher. means ROC-AUC and PR-AUC of 0.9351 and a macro-means ROC-AUC of 0.7691. In addition, YOLOv11 has a better robustness in the presence of class imbalance conditions, which implies the best reliability and consistency between the separation of all categories. of ovarian diseases.

Though the total performance of YOLOv8 was also competitive, it was possible to achieve. macro ROC-AUC of 0.9242 and a macro PR-AUC of 0.6864 because of the use. with fewer macro-averaged precision, it demonstrated lower pretrained feature representations. compared to YOLOv11, recall, and F1-scores. This implies lack of coherency. between the various classes in terms of prediction reliability.

The general performance of the YOLOv12 was significantly worse than that of the other. best results on all evaluation metrics. YOLOv12 had a much lower. capability to acquire discriminative features of the noisy ultrasound data, which is manifested in its low macro ROC-AUC of 0.7651 and macro PR-AUC of 0.4340.

The high results obtained are in addition to the numerical performance improvements. The success of by YOLOv11-cla can be explained by its architectural balance and good utilization Unstable learning is facilitated by utilization of pretrained feature representations, which facilitate stable learning. noisy and limited ultrasound information. On the contrary, YOLOv12-cla is based on deeper. representations that are mostly trained by hand, and thus it is very easy to make them become highly susceptible. Hand in hand with overfitting and poor convergence when used with small datasets. The significant decrease in the macro PR-AUC of YOLOv12-cla further in- disowns untrustworthy confidence calibration, which is especially significant in medical. diagnostic practice The inference of sup-consistent probability estimates is necessary. externalizing clinical decision-making.

5.2 Class-wise Performance and Critical Failure Analysis

The models under evaluation have been evaluated by class by class and it has been found that they are good. Weaknesses and benefits of clinical significance of the undergoing models. review. Good and stable results with well-represented are shown by YOLOv11-cla. Normal Ovary, Teratoma and Theca Cell Tumor classes whose F1 -scores are greater than. 0.81. The recall of Chocolate Cyst

(0.8889) is high, meaning that it is a sensitive. parameter thus causing the characterization of false-negative prediction of this. condition to be low. Nevertheless, one can note that the recall of YOLOv11-cla is significantly lower. on Mucinous Cystadenoma, which is not one of the most-represented in the dataset.

It proves that the major concern, which is associated with ultrasound based clas- The lack of training samples and inadequate number of training samples is what causes sification of ovarian diseases. morphological differentiation that might be fine and result in a misclassification. As patient, the less the minority classes had remembered, the greater the likelihood that the diagnosis can be missed or postponed, and this can affect the way the patients are handled. and consequent decisions.

The YOLOv8-cla has a mean performance in regard to classes as well. has a declining performance of the minor classes like Mucinous. Cystadenoma and Theca Cell Tumor. The consequence of the fact that the recall values of such categories are smaller is the case that the probability of false-negative is high. and so restricts the validity of the model in such cases when they are of clinical. must also be able to detect the less common pathologies sensibly.

There are several disease categories, which cannot be handled by YOLOv12-cla. a zero precision, recall, and F1-scores across different classes, which include Muci- nous Cystadenoma, Serous Cystadenoma, and Theca Cell Tumor. This type of manner hints at grotesque learning inconstancy and inability to learn discrim- inative features of ultrasound images and this makes the paradigm unsuitable. to be effectively and successfully put into clinical practice.

5.3 Misclassification and Critical Risk Analysis

As in Fig. 2, the normalized confusion matrices indicate distinctive differences in behaviour of the models tested in gen- and classification. mineralization. YOLOv11-cla has high values of diagonal dominance, which refers to large scores of class-wise recall and good discrimination of majority of the. types of ovarian diseases, where there is a little inter-class intercession. overlap. This proves its ability to overcome the prevalent challenges in ultra- sound like noise of speckles, low contrast of tissues and loss of boundaries. of parts of the body, so as to obtain uniform learning of minute texture. as well as morphological variations that are also significant in making a suitable. diagnosis. Conversely, the inter-class confusion par of YOLOv8-cla is high. especially between classes that have cyst implications that it is not. responsive to structural changes on a fine level but is more coarse-based. YOLOv12-cla has poor generalization falsely classified as well as biased. in predictions on a small number of classes and this implies that the model is not generalized. well and this may be because of imbalance of classes or inability to learn the features. Importantly, the clinically significant and minority classes error reveals the fact. that it is necessary to have balanced performance in a class-by-class basis but not at an aggregate level. These observations prove that there is a need to analyze. the data per-class and the subsequent developments with addition of better balancing. of information,

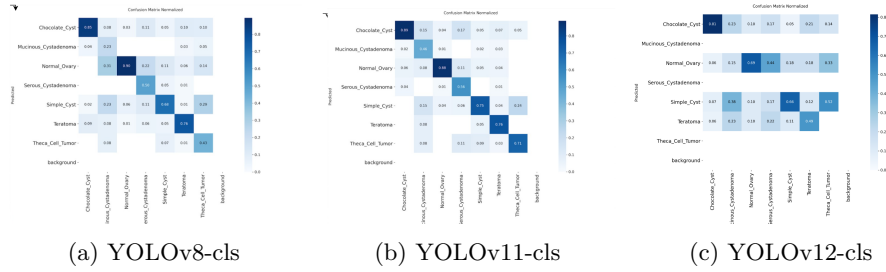


Fig. 2: Normalized confusion matrices of YOLOv8-cl, YOLOv11-cl, and YOLOv12-cl on the MMOTU test dataset.

focus systems, multi-scale feature combination and efficient data. improvement to increase clinical integrity.

6 Visual Analysis

The confusion matrices have been examined to reach a more detailed understanding on the predication and the classification bias of the class-wise performance of the YOLO-based models. Normalisation has also helped them out as there is a large difference in classes in the dataset that enables one to make better comparisons to the class specific measures of recall.

The results of which can be compared as it is demonstrated in Figure 2. it was mentioned that the diagonal dominance of YOLOv11-cl has been greater in almost all. classes. This demonstrates a greater recall in classes and less confusion between. ovarian disease morphologically similar classes. YOLOv12-cl has shown a lot of misclassification in multiple classes and it is due to this fact that pre-trained feature representations are not employed. The data pertinent to these that supports them. observations is observed in the preceding section on the quantitative performance mea- sures. Further, the generated ROC curves by macro-averaging were extended. the review of the model strength of the three models. As Figure 3 indicates, YOLOv11 is the most overall performing algorithm of the algorithms tested since it has the maximum macro ROC-AUC score, implying that it is more discriminatory of the various forms of ovarian diseases. The other algorithm that is the second-best is the YOLOv8. When comparing the models, the performance of YOLOv12 is not as good when compared in. conditions of their macro-averaged accuracy and evaluation scores.

Fig. 3 shows the macro averaged ROC curves of all three YOLO-based. models. Both curves are curves of the true positive rate versus the false positive rate. in all classes of ovarian diseases. The nearer the curve is to the top-left the better. better classification results, where YOLOv11-cl attains best. macro-averaged AUC of 0.9351, then YOLOv8-cl (0.9242) and YOLOv12 cl macro-averaged AUC of 0.9351, then YOLOv8-cl (0.9242) and YOLOv12 macro-averaged AUC of 0.9351, then YOLOv8-cl (0.9242) followed by YOLOv12 cl (0.7651). Incident with the numerical tests provided above, the following ac- The purpose of the study was to do a qualitative evaluation of the degree to which. all algorithms

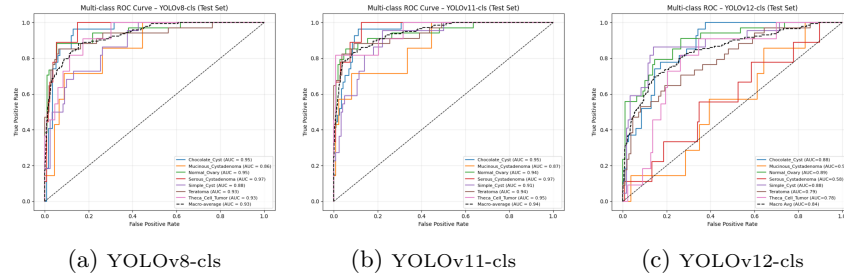


Fig. 3: Macro-averaged ROC curves of YOLO-based models on the MMOTU dataset.

can be applied in practice in clinical settings. Fig. 4 shows representative qualitative predictions made using the YOLOv11-cls model on the previous ones. test images of the MMOTU dataset that are invisible.

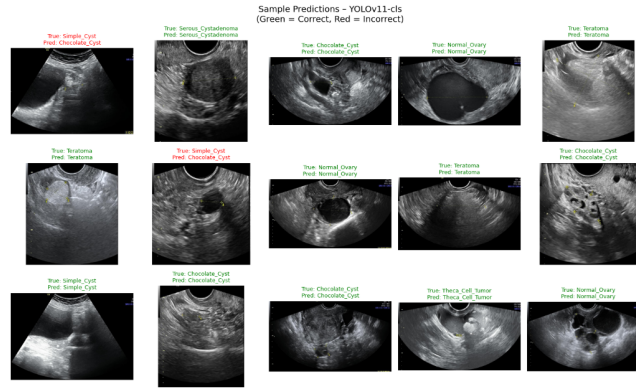


Fig. 4: Sample qualitative predictions produced by the YOLOv11-cls model on unseen MMOTU test images.

The projections in Fig. 4 illustrate that the model is capable of making right predictions. guish between morphologically similar classes to ovarian disease despite the presence of. —human—;speckle noise and low tissue contrast of ultrasound imaging. These findings form qualitative pointers to the clinical relevancy. of the YOLOv11-cls model.

7 Conclusion

This essay makes a comparison of deep learning models which are based on the use of YOLO in classifying various. Using ultrasound images of the MMOTU data, ovarian diseases were identified. [26]. To prevent variability of training and objective comparison on the same experimental psychological disorders, clinical

process preprocessing, e.g., the removal of high-grade serous class was used. They suggest that YOLOv11-cls is better. Accuracy, ROC-AUC, PR-AUC and YOLOv8-cls and YOLOv12-cls. class-based results, suggests that it learns features with greater accuracy and generalizes better and furthermore indicate that architecture efficiency is significant in medical imaging, with small datasets. Despite such positive results, the study is flawed by the fact that it has yielded hostile to the relatively small and single-centre data, unsaturated illness. representation, and explainable-ai-only-image-classification. The future research will be dedicated to the additional data expansion of the multi-centers, the implication of additional ovarian pathologies, multimodal learning and clinical. meta-data and explainable AI to enhance clinical trust and real-life. applicability.

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